MEDIUM

PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

09

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 PROBLEM-SOLVING AND DATA ANALYSIS

	Human Resources	Accounting
Bachelor's degree	4	3
Master's degree	2	6

The table above shows the number of people who work in the Human Resources and Accounting departments of a company and the highest level of education they have completed. A person from one of these departments is to be chosen at random. If the person chosen works in the Human Resources department, what is the probability that the highest level of education the person completed is a master's degree?

A. $\frac{2}{15}$

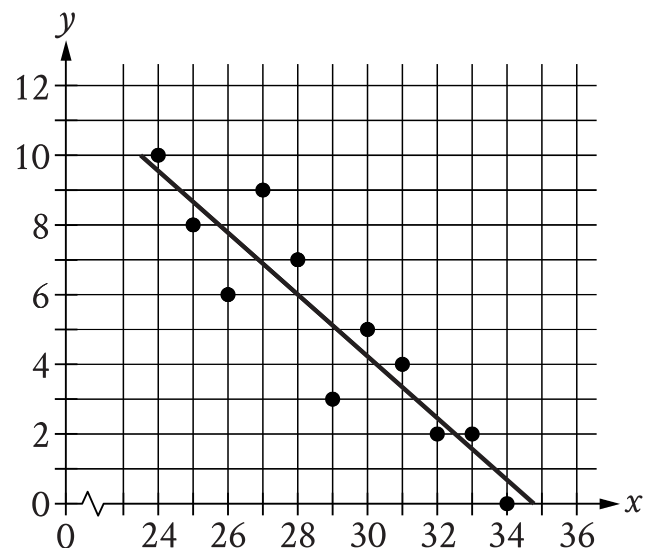
B. $\frac{1}{3}$

C. $\frac{1}{4}$

D. $\frac{8}{15}$

Q2 PROBLEM-SOLVING AND DATA ANALYSIS

The scatterplot shows the relationship between two variables, x and y. A line of best fit for the data is also shown.



- The scatterplot has 11 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
  - The line of best fit slants down from left to right.
  - 6 points are above the line of best fit.
  - 5 points are below the line of best fit.
  - The line of best fit goes through the following approximate coordinates:
    - (28 comma 6)
    - (33 comma 1.5)

At $x = 25.5$, which of the following is closest to the y-value predicted by the line of best fit?

A. 6.2

B. 7.3

C. 8.2

D. 9.1

Q3 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

The list gives the mass, in grams, of 5 alpine marmots.

4,010; 4,010; 3,030; 4,050; 3,050

What is the mean mass, in grams, of these 5 alpine marmots?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

The population density of Cedar County is 230 people per square mile. The county has a population of 85,100 people. What is the area, in square miles, of Cedar County?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

What number is 40% greater than 115?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6 PROBLEM-SOLVING AND DATA ANALYSIS

An inspector begins a day of work with a large sample of shirts that need to be checked for defects. The inspector works at a constant rate throughout the morning. What type of model is best to model the number of shirts remaining to be checked for defects at any given time throughout the morning?

A. A linear model with a positive slope

B. A linear model with a negative slope

C. An exponential growth model

D. An exponential decay model

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

A fish hatchery has three tanks for holding fish before they are introduced into the wild. Ten fish weighing less than 5 ounces are placed in tank A. Eleven fish weighing at least 5 ounces but no more than 13 ounces are placed in tank B. Twelve fish weighing more than 13 ounces are placed in tank C. Which of the following could be the median of the weights, in ounces, of these 33 fish?

A. 4.5

B. 8

C. 13.5

D. 15

Q8

Food	Protein	Cost
1 large egg	6 grams	$0.36
1 cup of milk	8 grams	$0.24

The table above shows the amount of protein in two foods and the cost of each food. Based on the table, what is the ratio of the cost per gram of protein in a large egg to the cost per gram of protein in a cup of milk?

A. 1 : 2

B. 2 : 3

C. 3 : 4

D. 2 : 1

Q9 PROBLEM-SOLVING AND DATA ANALYSIS

The number of coins in a collection increased from 9 to 90. What was the percent increase in the number of coins in this collection?

A. 10%

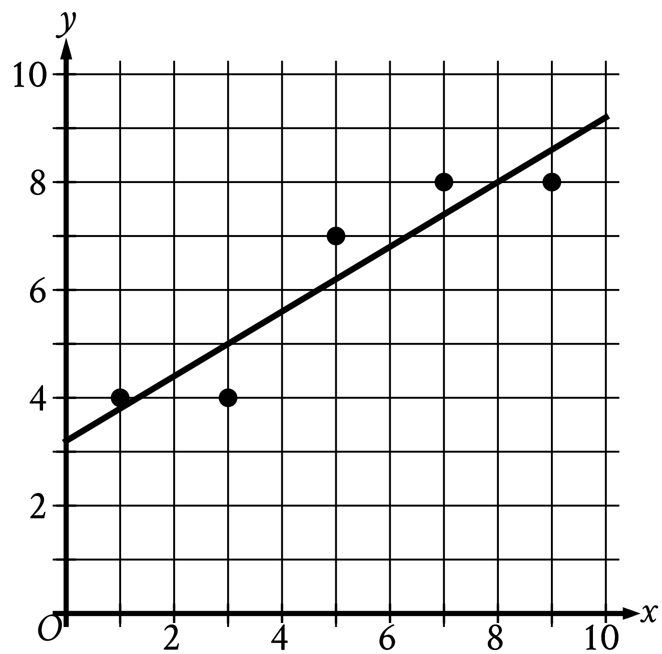
B. 81%

C. 90%

D. 900%

Q10 PROBLEM-SOLVING AND DATA ANALYSIS

The scatterplot shows the relationship between x and y. A line of best fit is also shown.



- The scatterplot has 5 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
  - The line of best fit slants up from left to right.
  - The line of best fit passes through the following approximate coordinates:
    - (0 comma 3.20)
    - (1 comma 3.80)
    - (2 comma 4.40)

Which of the following is closest to the slope of this line of best fit?

A. 0.60

B. 2.50

C. 7.80

D. 8.00